Jimmy Lee 李兆康

jimmylee@u.northwestern.edu · jimmylee728.github.io

ACADEMIC POSITIONS

Postdoctoral Fellow in Economics Sept 2024 – Present
Global Poverty Research Lab, Northwestern University

Visiting Scholar Jan 2025 – June 2026
Center for International Development, Harvard University

RESEARCH FIELDS

Development Economics, Program Evaluation
Focus—Synergies between Agriculture, Rural Education, and Digital Technologies

Family Economics, Organizational Economics
Focus—Intergenerational Dynamics in Rural Household and Community Decision Making

Behavioral Economics
Focus—The Economics of Curiosity: Endogenous Model Uncertainty under Cognitive Limits

EDUCATION

Ph.D., Economics, Northwestern University 2024
M.Phil., Economics, The Chinese University of Hong Kong 2017
B.SSc., Economics, The Chinese University of Hong Kong 2015

JOB MARKET PAPER

"Miscategorization as an Information Friction? Activating School-Based Agricultural Education in Liberia with Parental Engagement"

(Funded by AFD Fund for Innovation in Development, USAID-DIV, Wellspring Foundation, and FAO)

Many developing countries have under-staffed agricultural extension systems, while rural households face tradeoffs between children's schooling and farm labor. This paper provides a proof of concept that schools can become cost-effective hubs for these dual challenges, but only if parent-school linkages are strengthened. In a three-year Liberian experiment across 197 schools, I evaluate school-based agricultural education (SBAE)—a holistic model that trains science teachers to integrate inquiry-based learning with school demonstration farms and student home farming projects—estimating intent-to-treat effects on voluntary participants. I cross-randomize whether to inform parents about this integrated model via promotional videos and field days. SBAE improves targeted outcomes when paired with parental engagement: students' households adopt 26–53% more agricultural techniques, dropouts fall 4–5 percentage points per year, absenteeism falls 28%, and studying hours rise 22%. It is among the most cost-effective in agricultural extension trials, and over twice as cost-effective for school attendance as a conditional cash transfer benchmark (Baird et al. 2011). SBAE's positive training effects on teachers and students are unaltered by parental engagement. However, without parental engagement, SBAE's diffusion mechanisms—school demonstration farms, home projects, intra-household communication—stay muted. Technology adoption effects vanish, absenteeism rises 31%, studying hours fall 20%, and parents disconnect—reducing school farm visits 12% and parent–teacher meetings 39%. Parents' persistent avoidance of information, plus sharply reversed responses under engagement, pose a puzzle. Bayesian updating, selective attention, and associative memory each fall short alone. I propose an integrative explanation: parental engagement shifts parents away from viewing agriculture and education as substitutes.

WORKING PAPERS

1. **"Technology Adoption and Schooling as Complements: The Promise of School-Based Agricultural Education in Africa"** (joint with Christopher Udry)

   In sub-Saharan Africa, wide heterogeneity across farms makes other households' experience a noisy guide to what suits a household's own farms, slowing technology diffusion, while low perceived returns to schooling drive students to drop out. We develop a model of why school-based agricultural education (SBAE), a holistic program in which trained science teachers integrate inquiry-based learning with

school demonstration farms and student home farming projects, addresses this dual problem. Our model builds on the target-input framework, in which a household learns a technology's suitability only by using it in production or by observing others who do. We add a third channel: the more of a child's time the household devotes to schooling, the more precisely it learns from the school's production signals. Schooling and technology adoption become complements. First, schooling carries an option value on the farm: by raising the probability of future adoption, it makes attendance worthwhile even where labor-market returns are low. Second, SBAE effectively subsidizes information acquisition by delivering information through schooling, whose cost households already bear for its educational returns, countering the under-supply of information in the presence of learning spillovers. We provide experimental evidence from Liberia that SBAE cost-effectively raises both agricultural technology adoption and school attendance, and use extensions of the model to study how SBAE's design can be strengthened, including (i) how effectively inquiry-based learning raises the marginal informational return to schooling, and (ii) how effectively SBAE's home visits aggregate households' dispersed knowledge and feed it back to them.

2. **"Endogenous Model Uncertainty and Curiosity under Cognitive Limits"**

   Neuroscience suggests that people automatically classify a situation before choosing how to act. Association with experience supplies a familiar interpretation, but when it explains the evidence poorly, people begin to entertain less familiar alternatives. Model-switching theories capture this shift partially: they allow one model at a time and abstract from information acquisition. This paper treats contextual shifts in "mindsets" as a choice of how much model uncertainty to entertain. Each mental model pairs a prior with a view of which features reveal the state, so weighting models means choosing what to attend to. After an initial signal, an agent with a costless default and latent, less precise alternatives chooses these weights and how much information to acquire. The key discipline is a neurally grounded evidence constraint that bounds motivated beliefs: an alternative's weight is capped by its Bayesian posterior. Curiosity arises endogenously as a drive to resolve the resulting model uncertainty that imposes hedging costs. The model isolates new mechanisms for two effects central to behavioral models of associative memory. First, context effects: payoff-irrelevant context sways decisions only when another model makes it relevant, and the curiosity this creates drives self-correction. Persistent context effects thus reflect the information environment. Second, interference effects: rather than changing which model comes to mind, rival explanations dilute one another's support, leaving none activated and search dormant. I illustrate design principles for stimulating curiosity about new programs, pedagogies, and farming techniques using examples from development economics.

3. **"Can Communicating Parents' Beliefs to Students Be a Pivotal Component of Parental Engagement? Evidence from School-Based Agricultural Education in Liberia"**

   (Funded by NSF, Weiss Fund, Global Poverty Research Lab, and the Buffett Institute for Global Affairs)

   When school programs require students to introduce improved techniques on household farms, is informing students and parents about the program sufficient, or is there a case for further signaling parents' support to students? I study this question within a randomized evaluation of school-based agricultural education (SBAE) in Liberia, where students take school farm training into a home farming project. After parents view a promotional video of the program and students are encouraged to diffuse techniques to their parents, I randomize a nudge revealing to students their parents' optimism about their farm-management skills. Two years later, SBAE improves students' livelihoods only with the nudge: income rises 63%, on-time school fee payment 43–47%, and cost-effectiveness for attendance doubles. Analysis of mechanisms highlights a management aspect of agricultural technology adoption that can be addressed with nudges: even where parents become attentive to SBAE and adopt promoted techniques, third-party assurance can be pivotal in shifting students away from assisting in subsistence farming.

WORK IN PROGRESS

Ongoing Randomized Experiments (scaling SBAE across Liberia, Côte d'Ivoire, and Kenya in 2026-35)

1. **"Learning to Learn: Synergizing Inquiry-based Science and Agricultural Transformation in Liberia"** (joint with Vesall Nourani, Harvard GSE; scheduled RCT within the scaling of SBAE in 2026-29)
2. **"Parental Engagement in School-based Agricultural Education: Testing Mechanisms at Scale"** (piloted interventions in 2025-26)
3. **"Parental Engagement and the Use of Unconditional Cash Transfers"**
4. **"Directed Technical Change under Data-Driven Farming Advice: An Empirical Framework"** (grant from Weiss Fund; piloted survey method; planning RCT)

GRANTS AND AWARDS

Total program evaluation and academic research grants secured for SBAE in Liberia: **$4.2 million**

Program Evaluation Grants

AFD Fund for Innovation in Development ($1.2 mil; co-PI)	2025
J-PAL Learning for All Initiative: Scaling Award ($222K; co-PI)	2024
USAID Development Innovation Ventures ($395K; co-PI)	2022
Wellspring Foundation ($192K; co-PI)	2022
AFD Fund for Innovation in Development ($1.2 mil; co-PI)	2021
Food and Agriculture Organization ($68K; co-PI)	2021

Academic Research Awards

J-PAL Learning for All Initiative ($400K; co-PI)	2025
Weiss Fund ($5K; Lead PI)	2022
National Science Foundation ($451K; co-PI)	2020
Weiss Fund ($29K; Lead PI)	2020
Global Poverty Research Lab ($19K; Lead PI)	2019
Buffett Institute for Global Affairs ($5K; Lead PI)	2019

INVITED PRESENTATIONS

During postdoctoral studies:

National Chung Cheng University; NYU Africa House–Center for Technology, Economics, and Development (CTED); Mohammed VI Polytechnic University (UM6P); Tel Aviv University; Oklahoma State University; Harvard Center for International Development; Africa Food Systems Forum 2024

During PhD:

Meeting of the Society of Economics of the Household (SEHO); J-PAL—Middle East and North Africa (MENA); CGIAR—Standing Panel on Impact Assessment (SPIA); International Food Policy Research Institute (IFPRI); University of Glasgow; National Graduate Institute for Policy Studies (GRIPS, Tokyo); Development Day 2023

TEACHING

Teaching Assistant, Northwestern University (Department median 5.3/6)

Introduction to Microeconomics (5.2, 5.4, 5.6), Introduction to Macroeconomics (5.4, 5.4), Intermediate Microeconomics (5.4), Intermediate Macroeconomics (5.7), International Trade (5.6), Economics of Nonprofit Organizations (5.8)

Teaching Assistant, The Chinese University of Hong Kong

Mathematical Methods in Economics I, II, Family Economics, International Macroeconomics

LANGUAGES AND SKILLS

Citizenship: Hong Kong Special Administrative Region, China
Languages: English (fluent), French (intermediate), Mandarin (fluent), Cantonese (native)
Programming: Stata, R, Matlab, Python, ArcGIS, SurveyCTO

REFERENCES

Professor Christopher Udry (Chair)
Department of Economics
Northwestern University
2211 Campus Drive, Evanston, IL 60208
(+1) 847-491-8216
christopher.udry@northwestern.edu

Professor Dean Karlan
Department of Economics
Northwestern University
2211 Campus Drive, Evanston, IL 60208
(+1) 847-491-8706
karlan@u.northwestern.edu

Professor Lori Beaman
Department of Economics
Northwestern University
2211 Campus Drive, Evanston, IL 60208
(+1) 847-491-5394
l-beaman@northwestern.edu